In [1]:
```
import optuna
from time import time
import numpy as np
import pandas as pd

from sklearn.model_selection import train_test_split
from sklearn.model_selection import cross_val_score
from sklearn.linear_model import LogisticRegression
from sklearn.svm import SVC
from sklearn.svm import SVR
from sklearn.kernel_ridge import KernelRidge
from sklearn.gaussian_process import GaussianProcessRegressor

from sklearn.gaussian_process.kernels import RBF, Matern, DotProduct, WhiteKernel

from sklearn.metrics import mean_squared_error
from sklearn.metrics import r2_score
from sklearn.metrics import mean_absolute_error

from sklearn.metrics import accuracy_score
from sklearn.metrics import f1_score
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay

from sklearn.preprocessing import StandardScaler
import matplotlib.pyplot as plt
```

In [2]:
```
df = pd.read_csv("winequality-red.csv")
df.head()
```

Out[2]:

	fixed acidity	volatile acidity	citric acid	residual sugar	chlorides	free sulfur dioxide	total sulfur dioxide	density	pH	sulphates	a
0	7.4	0.70	0.00	1.9	0.076	11.0	34.0	0.9978	3.51	0.56	
1	7.8	0.88	0.00	2.6	0.098	25.0	67.0	0.9968	3.20	0.68	
2	7.8	0.76	0.04	2.3	0.092	15.0	54.0	0.9970	3.26	0.65	
3	11.2	0.28	0.56	1.9	0.075	17.0	60.0	0.9980	3.16	0.58	
4	7.4	0.70	0.00	1.9	0.076	11.0	34.0	0.9978	3.51	0.56	

In [3]:
```
X = df.drop("quality", axis=1)
y = df["quality"]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.3, stratify=y

from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
```

```
# Check shapes
print("Train data shape:", X_train.shape)
print("Test data shape:", X_test.shape)
```

```
Train data shape: (1119, 11)
Test data shape: (480, 11)
```

In [4]:

```
#for letter a
def create_model(trial):
    model_type = trial.suggest_categorical('model_type',['Logistic-regression', 'SV

    if model_type == 'SVM':
        kernel = trial.suggest_categorical('SVM-kernel',['linear','poly','rbf','sig
        C = trial.suggest_float('SVM-C',0.1, 1000,log=True)
        gamma = trial.suggest_float('SVM-gamma',0.0001, 10,log=True)
        model = SVC(kernel=kernel,C=C,gamma=gamma)

    if model_type == 'Logistic-regression':
        penalty = trial.suggest_categorical('penalty',['l1','l2'])
        if penalty == 'l1':
            solver = 'saga'
        else:
            solver = 'lbfgs'
        regularization = trial.suggest_float('Logistic-regularization',0.01, 100,lo
        model = LogisticRegression(penalty=penalty,C=regularization,solver=solver,m

    if trial.should_prune():
        raise optuna.TrialPruned()

    return model
```

In [5]:

```
def objective(trial):
    model = create_model(trial)
    scores = cross_val_score(model,X_train,y_train,cv=5,scoring="accuracy")
    return np.mean(scores)
```

In [6]:

```
start = time()

study = optuna.create_study(direction='maximize')
study.optimize(objective, n_trials=100)

best_params = study.best_params

print("\nBest parameters:", best_params)
```

```
[I 2026-03-06 19:37:25,002] A new study created in memory with name: no-name-80340b5
a-5836-43de-832b-458db83e33a6
[I 2026-03-06 19:37:25,302] Trial 0 finished with value: 0.5844650864830238 and para
meters: {'model_type': 'Logistic-regression', 'penalty': 'l2', 'Logistic-regularizat
ion': 29.103898700661155}. Best is trial 0 with value: 0.5844650864830238.
[I 2026-03-06 19:37:25,501] Trial 1 finished with value: 0.5773382447149263 and para
meters: {'model_type': 'SVM', 'SVM-kernel': 'rbf', 'SVM-C': 60.97277716904934, 'SVM-
gamma': 0.0001729966039399193}. Best is trial 0 with value: 0.5844650864830238.
[I 2026-03-06 19:37:27,120] Trial 2 finished with value: 0.5844610826393337 and para
meters: {'model_type': 'Logistic-regression', 'penalty': 'l1', 'Logistic-regularizat
ion': 1.4984472184141848}. Best is trial 0 with value: 0.5844650864830238.
[I 2026-03-06 19:37:27,283] Trial 3 finished with value: 0.5853579436258809 and para
meters: {'model_type': 'Logistic-regression', 'penalty': 'l2', 'Logistic-regularizat
ion': 2.1125205021632563}. Best is trial 3 with value: 0.5853579436258809.
[I 2026-03-06 19:37:27,340] Trial 4 finished with value: 0.5853579436258809 and para
meters: {'model_type': 'Logistic-regression', 'penalty': 'l2', 'Logistic-regularizat
ion': 0.032014971670809154}. Best is trial 3 with value: 0.5853579436258809.
[I 2026-03-06 19:37:27,651] Trial 5 finished with value: 0.5853579436258809 and para
meters: {'model_type': 'Logistic-regression', 'penalty': 'l1', 'Logistic-regularizat
ion': 0.24593874633533275}. Best is trial 3 with value: 0.5853579436258809.
[I 2026-03-06 19:37:29,811] Trial 6 finished with value: 0.5808856502242152 and para
meters: {'model_type': 'Logistic-regression', 'penalty': 'l1', 'Logistic-regularizat
ion': 2.4391156212994165}. Best is trial 3 with value: 0.5853579436258809.
[I 2026-03-06 19:37:34,991] Trial 7 finished with value: 0.5844650864830238 and para
meters: {'model_type': 'Logistic-regression', 'penalty': 'l1', 'Logistic-regularizat
ion': 14.225976970521925}. Best is trial 3 with value: 0.5853579436258809.
[I 2026-03-06 19:37:35,340] Trial 8 finished with value: 0.5835722293401666 and para
meters: {'model_type': 'Logistic-regression', 'penalty': 'l2', 'Logistic-regularizat
ion': 65.08742062457688}. Best is trial 3 with value: 0.5853579436258809.
[I 2026-03-06 19:37:35,592] Trial 9 finished with value: 0.5844650864830238 and para
meters: {'model_type': 'Logistic-regression', 'penalty': 'l2', 'Logistic-regularizat
ion': 20.57209241605416}. Best is trial 3 with value: 0.5853579436258809.
[I 2026-03-06 19:37:35,708] Trial 10 finished with value: 0.581806534272902 and para
meters: {'model_type': 'SVM', 'SVM-kernel': 'linear', 'SVM-C': 0.13219343255764254,
'SVM-gamma': 7.310144444763054}. Best is trial 3 with value: 0.5853579436258809.
[I 2026-03-06 19:37:36,545] Trial 11 finished with value: 0.5826513452914799 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 581.8174940211265, 'SV
M-gamma': 0.05771873953838909}. Best is trial 3 with value: 0.5853579436258809.
[I 2026-03-06 19:37:36,590] Trial 12 finished with value: 0.5835802370275465 and par
ameters: {'model_type': 'Logistic-regression', 'penalty': 'l2', 'Logistic-regulariza
tion': 0.010385792769858914}. Best is trial 3 with value: 0.5853579436258809.
[I 2026-03-06 19:37:36,652] Trial 13 finished with value: 0.5898302370275464 and par
ameters: {'model_type': 'Logistic-regression', 'penalty': 'l2', 'Logistic-regulariza
tion': 0.07529000161012195}. Best is trial 13 with value: 0.5898302370275464.
[I 2026-03-06 19:37:36,854] Trial 14 finished with value: 0.4262732222934017 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'sigmoid', 'SVM-C': 0.3381690027623724
4, 'SVM-gamma': 0.00012384605186296265}. Best is trial 13 with value: 0.589830237027
5464.
[I 2026-03-06 19:37:36,938] Trial 15 finished with value: 0.5862508007687379 and par
ameters: {'model_type': 'Logistic-regression', 'penalty': 'l2', 'Logistic-regulariza
tion': 0.2519266918040795}. Best is trial 13 with value: 0.5898302370275464.
[I 2026-03-06 19:37:37,021] Trial 16 finished with value: 0.5898302370275464 and par
ameters: {'model_type': 'Logistic-regression', 'penalty': 'l2', 'Logistic-regulariza
tion': 0.1968022803830004}. Best is trial 13 with value: 0.5898302370275464.
[I 2026-03-06 19:37:37,092] Trial 17 finished with value: 0.5898302370275464 and par
ameters: {'model_type': 'Logistic-regression', 'penalty': 'l2', 'Logistic-regulariza
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tion': 0.12242368448219598}. Best is trial 13 with value: 0.5898302370275464.
[I 2026-03-06 19:37:37,251] Trial 18 finished with value: 0.4334881486226777 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'sigmoid', 'SVM-C': 4.016149385464848,
'SVM-gamma': 8.96994829602774}. Best is trial 13 with value: 0.5898302370275464.
[I 2026-03-06 19:37:37,313] Trial 19 finished with value: 0.5907270980140936 and par
ameters: {'model_type': 'Logistic-regression', 'penalty': 'l2', 'Logistic-regulariza
tion': 0.05954210808952923}. Best is trial 19 with value: 0.5907270980140936.
[I 2026-03-06 19:37:37,367] Trial 20 finished with value: 0.5889373798846893 and par
ameters: {'model_type': 'Logistic-regression', 'penalty': 'l2', 'Logistic-regulariza
tion': 0.04617279249776519}. Best is trial 19 with value: 0.5907270980140936.
[I 2026-03-06 19:37:37,432] Trial 21 finished with value: 0.5898302370275463 and par
ameters: {'model_type': 'Logistic-regression', 'penalty': 'l2', 'Logistic-regulariza
tion': 0.07948427190452056}. Best is trial 19 with value: 0.5907270980140936.
[I 2026-03-06 19:37:37,534] Trial 22 finished with value: 0.5835682254964766 and par
ameters: {'model_type': 'Logistic-regression', 'penalty': 'l2', 'Logistic-regulariza
tion': 0.45232320181326635}. Best is trial 19 with value: 0.5907270980140936.
[I 2026-03-06 19:37:37,589] Trial 23 finished with value: 0.5844770980140936 and par
ameters: {'model_type': 'Logistic-regression', 'penalty': 'l2', 'Logistic-regulariza
tion': 0.014279175051804654}. Best is trial 19 with value: 0.5907270980140936.
[I 2026-03-06 19:37:37,699] Trial 24 finished with value: 0.5808856502242152 and par
ameters: {'model_type': 'Logistic-regression', 'penalty': 'l2', 'Logistic-regulariza
tion': 0.6726705150805666}. Best is trial 19 with value: 0.5907270980140936.
[I 2026-03-06 19:37:37,769] Trial 25 finished with value: 0.5898302370275464 and par
ameters: {'model_type': 'Logistic-regression', 'penalty': 'l2', 'Logistic-regulariza
tion': 0.12081424640388627}. Best is trial 19 with value: 0.5907270980140936.
[I 2026-03-06 19:37:37,825] Trial 26 finished with value: 0.5835682254964766 and par
ameters: {'model_type': 'Logistic-regression', 'penalty': 'l2', 'Logistic-regulariza
tion': 0.030490021349073988}. Best is trial 19 with value: 0.5907270980140936.
[I 2026-03-06 19:37:38,230] Trial 27 finished with value: 0.5782391095451633 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'linear', 'SVM-C': 7.593527376210658,
'SVM-gamma': 0.027010464685253886}. Best is trial 19 with value: 0.5907270980140936.
[I 2026-03-06 19:37:38,540] Trial 28 finished with value: 0.5835722293401666 and par
ameters: {'model_type': 'Logistic-regression', 'penalty': 'l1', 'Logistic-regulariza
tion': 0.21184966130886013}. Best is trial 19 with value: 0.5907270980140936.
[I 2026-03-06 19:37:38,607] Trial 29 finished with value: 0.5916199551569508 and par
ameters: {'model_type': 'Logistic-regression', 'penalty': 'l2', 'Logistic-regulariza
tion': 0.053241843266762914}. Best is trial 29 with value: 0.5916199551569508.
[I 2026-03-06 19:37:38,674] Trial 30 finished with value: 0.5916199551569508 and par
ameters: {'model_type': 'Logistic-regression', 'penalty': 'l2', 'Logistic-regulariza
tion': 0.05057215977387461}. Best is trial 29 with value: 0.5916199551569508.
[I 2026-03-06 19:37:38,737] Trial 31 finished with value: 0.5889413837283792 and par
ameters: {'model_type': 'Logistic-regression', 'penalty': 'l2', 'Logistic-regulariza
tion': 0.06455207232110727}. Best is trial 29 with value: 0.5916199551569508.
[I 2026-03-06 19:37:38,792] Trial 32 finished with value: 0.5853539397821909 and par
ameters: {'model_type': 'Logistic-regression', 'penalty': 'l2', 'Logistic-regulariza
tion': 0.024787542028321542}. Best is trial 29 with value: 0.5916199551569508.
[I 2026-03-06 19:37:38,851] Trial 33 finished with value: 0.5889413837283792 and par
ameters: {'model_type': 'Logistic-regression', 'penalty': 'l2', 'Logistic-regulariza
tion': 0.019293109122906812}. Best is trial 29 with value: 0.5916199551569508.
[I 2026-03-06 19:37:38,910] Trial 34 finished with value: 0.5916199551569508 and par
ameters: {'model_type': 'Logistic-regression', 'penalty': 'l2', 'Logistic-regulariza
tion': 0.055236899754040635}. Best is trial 29 with value: 0.5916199551569508.
[I 2026-03-06 19:37:38,975] Trial 35 finished with value: 0.5880365150544523 and par
ameters: {'model_type': 'Logistic-regression', 'penalty': 'l2', 'Logistic-regulariza
tion': 0.039140219339576475}. Best is trial 29 with value: 0.5916199551569508.
[I 2026-03-06 19:37:39,133] Trial 36 finished with value: 0.5951913837283793 and par
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ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 819.2474722053153, 'SV
M-gamma': 0.009358098799729108}. Best is trial 36 with value: 0.5951913837283793.
[I 2026-03-06 19:37:39,283] Trial 37 finished with value: 0.5907311018577834 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 571.3959542536347, 'SV
M-gamma': 0.008389643627787957}. Best is trial 36 with value: 0.5951913837283793.
[I 2026-03-06 19:37:39,443] Trial 38 finished with value: 0.42448750800768736 and pa
rameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 57.05738800280291, 'S
VM-gamma': 0.0026496055726684207}. Best is trial 36 with value: 0.5951913837283793.
[I 2026-03-06 19:37:39,738] Trial 39 finished with value: 0.5862267777065983 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'rbf', 'SVM-C': 81.68558011034479, 'SVM
-gamma': 0.38635392839708826}. Best is trial 36 with value: 0.5951913837283793.
[I 2026-03-06 19:37:39,871] Trial 40 finished with value: 0.4262732222934017 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 1.434150486831369, 'SV
M-gamma': 0.0017891109057071894}. Best is trial 36 with value: 0.5951913837283793.
[I 2026-03-06 19:37:40,027] Trial 41 finished with value: 0.5934016655989749 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 983.6671616930715, 'SV
M-gamma': 0.0072911239551986685}. Best is trial 36 with value: 0.5951913837283793.
[I 2026-03-06 19:38:00,508] Trial 42 finished with value: 0.5665999359385009 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 754.4679583679026, 'SV
M-gamma': 0.19930667563355764}. Best is trial 36 with value: 0.5951913837283793.
[I 2026-03-06 19:38:00,646] Trial 43 finished with value: 0.4262732222934017 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 172.9628577666554, 'SV
M-gamma': 0.0011040098564571573}. Best is trial 36 with value: 0.5951913837283793.
[I 2026-03-06 19:38:00,788] Trial 44 finished with value: 0.5916279628443306 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 201.67830555436345, 'S
VM-gamma': 0.011757962997464646}. Best is trial 36 with value: 0.5951913837283793.
[I 2026-03-06 19:38:00,931] Trial 45 finished with value: 0.5987668161434978 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 220.63286740100006, 'S
VM-gamma': 0.0126491782634641}. Best is trial 45 with value: 0.5987668161434978.
[I 2026-03-06 19:38:01,081] Trial 46 finished with value: 0.6023422485586163 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 229.23948797075852, 'S
VM-gamma': 0.014805768577832161}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:38:01,226] Trial 47 finished with value: 0.5933976617552851 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 217.9047523124353, 'SV
M-gamma': 0.012208473989189776}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:38:01,374] Trial 48 finished with value: 0.5889413837283792 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 905.782507987175, 'SVM
-gamma': 0.007118854520256456}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:38:01,649] Trial 49 finished with value: 0.5853419282511211 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 253.14111641821873, 'S
VM-gamma': 0.04016197775504408}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:38:01,796] Trial 50 finished with value: 0.5424847853939783 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 250.19963018347255, 'S
VM-gamma': 0.006338364496665456}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:38:01,943] Trial 51 finished with value: 0.5960962524023062 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 184.55944938923398, 'S
VM-gamma': 0.014792022742004815}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:38:02,117] Trial 52 finished with value: 0.5898062139654068 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 335.6945383103857, 'SV
M-gamma': 0.020274900304468223}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:38:02,256] Trial 53 finished with value: 0.43611467008328 and param
eters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 98.00581576837934, 'SVM-
gamma': 0.004601991792419282}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:38:02,489] Trial 54 finished with value: 0.5826633568225497 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 36.26872060729561, 'SV
M-gamma': 0.0649332569448859}. Best is trial 46 with value: 0.6023422485586163.
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[I 2026-03-06 19:38:02,659] Trial 55 finished with value: 0.5915959320948111 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 405.28881542851, 'SVM-
gamma': 0.01867668280273285}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:38:02,790] Trial 56 finished with value: 0.4262732222934017 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 972.2356171109195, 'SV
M-gamma': 0.000666242971671642}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:38:02,935] Trial 57 finished with value: 0.42448750800768736 and pa
rameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 23.47639537941864, 'S
VM-gamma': 0.004132034563926618}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:38:03,085] Trial 58 finished with value: 0.5978739590006407 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 150.96299536601487, 'S
VM-gamma': 0.014404358844025533}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:38:04,033] Trial 59 finished with value: 0.584437059577194 and para
meters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 142.1424399479795, 'SVM
-gamma': 0.09933961833956126}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:38:04,508] Trial 60 finished with value: 0.5799487508007688 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'rbf', 'SVM-C': 379.6468544015699, 'SVM
-gamma': 0.021210005280924718}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:38:04,655] Trial 61 finished with value: 0.5683896540679052 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 140.4089777270458, 'SV
M-gamma': 0.01052371748636326}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:38:04,810] Trial 62 finished with value: 0.5969650864830236 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 360.33967167789507, 'S
VM-gamma': 0.014141695977545185}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:38:05,041] Trial 63 finished with value: 0.5826633568225495 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 393.6884411738647, 'SV
M-gamma': 0.02977299040477535}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:38:05,187] Trial 64 finished with value: 0.4674087123638693 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 545.5613554196075, 'SV
M-gamma': 0.0037646454245558554}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:38:09,827] Trial 65 finished with value: 0.5755605381165919 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'linear', 'SVM-C': 104.5699110533164,
'SVM-gamma': 0.013172409228694393}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:38:09,976] Trial 66 finished with value: 0.4772141255605381 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'sigmoid', 'SVM-C': 540.3501602378916,
'SVM-gamma': 0.03925932985365471}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:38:10,496] Trial 67 finished with value: 0.5683736386931455 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'rbf', 'SVM-C': 951.4007976825853, 'SVM
-gamma': 0.0941495923159875}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:38:10,631] Trial 68 finished with value: 0.4262732222934017 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 47.934996030222095, 'S
VM-gamma': 0.002244717058586911}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:39:30,251] Trial 69 finished with value: 0.5755605381165919 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'linear', 'SVM-C': 308.6308896402704,
'SVM-gamma': 0.006419849708760875}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:39:30,435] Trial 70 finished with value: 0.4254043882126842 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'sigmoid', 'SVM-C': 500.98952888854194,
'SVM-gamma': 2.798213115797411}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:39:30,583] Trial 71 finished with value: 0.6014533952594491 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 205.22539334920984, 'S
VM-gamma': 0.01356734669350147}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:39:30,732] Trial 72 finished with value: 0.5978739590006407 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 147.99853071926267, 'S
VM-gamma': 0.014388500767537945}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:39:30,887] Trial 73 finished with value: 0.5969770980140936 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 130.6365782751438, 'SV
```

```
M-gamma': 0.017439628862360633}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:39:31,035] Trial 74 finished with value: 0.5942985265855221 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 97.48948790936704, 'SV
M-gamma': 0.01890019085749228}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:39:31,258] Trial 75 finished with value: 0.5826633568225497 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 144.43690994281624, 'S
VM-gamma': 0.040626583705255716}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:39:31,403] Trial 76 finished with value: 0.5943025304292121 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 72.79891869826166, 'SV
M-gamma': 0.016887452078804717}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:39:31,546] Trial 77 finished with value: 0.5934016655989749 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 17.936991747521404, 'S
VM-gamma': 0.027886386251800642}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:39:31,891] Trial 78 finished with value: 0.5817584881486227 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 139.9241576566231, 'SV
M-gamma': 0.0582466515734312}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:39:32,153] Trial 79 finished with value: 0.5782391095451633 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'linear', 'SVM-C': 4.002459698863834,
'SVM-gamma': 0.010229582011303855}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:39:32,342] Trial 80 finished with value: 0.4995675848814862 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'sigmoid', 'SVM-C': 219.78493085231668,
'SVM-gamma': 0.013969538568908655}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:39:32,486] Trial 81 finished with value: 0.5737628122998079 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 300.6143111609877, 'SV
M-gamma': 0.00847805369458957}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:39:32,628] Trial 82 finished with value: 0.45489670083279954 and pa
rameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 192.17327210493326,
'SVM-gamma': 0.0048937932046456255}. Best is trial 46 with value: 0.602342248558616
3.
[I 2026-03-06 19:39:32,780] Trial 83 finished with value: 0.43969010249839846 and pa
rameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 0.6459461215071678,
'SVM-gamma': 0.029215657574816322}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:39:32,935] Trial 84 finished with value: 0.5969851057014735 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 650.2101273841345, 'SV
M-gamma': 0.010766002242499472}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:39:33,068] Trial 85 finished with value: 0.4262732222934017 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 0.11950071142017926,
'SVM-gamma': 0.014573524453447572}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:39:33,215] Trial 86 finished with value: 0.5558816463805254 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'poly', 'SVM-C': 115.37734846550735, 'S
VM-gamma': 0.009287433924541188}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:39:33,455] Trial 87 finished with value: 0.5978819666880206 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'rbf', 'SVM-C': 201.7427872523066, 'SVM
-gamma': 0.003040751136424361}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:39:33,659] Trial 88 finished with value: 0.5970051249199232 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'rbf', 'SVM-C': 62.70164688312663, 'SVM
-gamma': 0.0033165734417892434}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:39:33,855] Trial 89 finished with value: 0.5862708199871877 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'rbf', 'SVM-C': 66.54535606064378, 'SVM
-gamma': 0.0013162871320505116}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:39:34,108] Trial 90 finished with value: 0.5925168161434977 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'rbf', 'SVM-C': 246.3203088177611, 'SVM
-gamma': 0.0031436218769689554}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:39:34,317] Trial 91 finished with value: 0.5862788276745675 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'rbf', 'SVM-C': 178.72560706000604, 'SV
M-gamma': 0.0009548813605819323}. Best is trial 46 with value: 0.6023422485586163.
```

```
[I 2026-03-06 19:39:34,765] Trial 92 finished with value: 0.599651665598975 and para
meters: {'model_type': 'SVM', 'SVM-kernel': 'rbf', 'SVM-C': 663.1224095505384, 'SVM-
gamma': 0.006095954251766063}. Best is trial 46 with value: 0.6023422485586163.
[I 2026-03-06 19:39:35,171] Trial 93 finished with value: 0.6076993914157592 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'rbf', 'SVM-C': 586.0532659005338, 'SVM
-gamma': 0.005111339524839813}. Best is trial 93 with value: 0.6076993914157592.
[I 2026-03-06 19:39:35,477] Trial 94 finished with value: 0.5934056694426649 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'rbf', 'SVM-C': 615.6691103339765, 'SVM
-gamma': 0.002280533117525094}. Best is trial 93 with value: 0.6076993914157592.
[I 2026-03-06 19:39:35,714] Trial 95 finished with value: 0.5844811018577836 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'rbf', 'SVM-C': 596.1458402728795, 'SVM
-gamma': 0.0004549602602393962}. Best is trial 93 with value: 0.6076993914157592.
[I 2026-03-06 19:39:35,912] Trial 96 finished with value: 0.5987708199871877 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'rbf', 'SVM-C': 44.123959485299316, 'SV
M-gamma': 0.005853177140256720 5}. Best is trial 93 with value: 0.6076993914157592.
[I 2026-03-06 19:39:36,115] Trial 97 finished with value: 0.5951993914157591 and par
ameters: {'model_type': 'SVM', 'SVM-kernel': 'rbf', 'SVM-C': 35.82674700465297, 'SVM
-gamma': 0.005711990560783214}. Best is trial 93 with value: 0.6076993914157592.
[I 2026-03-06 19:39:36,346] Trial 98 finished with value: 0.588961402946829 and para
meters: {'model_type': 'SVM', 'SVM-kernel': 'rbf', 'SVM-C': 83.58636266917262, 'SVM-
gamma': 0.0017882776531418874}. Best is trial 93 with value: 0.6076993914157592.
[I 2026-03-06 19:39:36,557] Trial 99 finished with value: 0.596108263933376 and para
meters: {'model_type': 'SVM', 'SVM-kernel': 'rbf', 'SVM-C': 52.33381247558941, 'SVM-
gamma': 0.00341411763750913}. Best is trial 93 with value: 0.6076993914157592.
Best parameters: {'model_type': 'SVM', 'SVM-kernel': 'rbf', 'SVM-C': 586.05326590053
38, 'SVM-gamma': 0.005111339524839813}
```

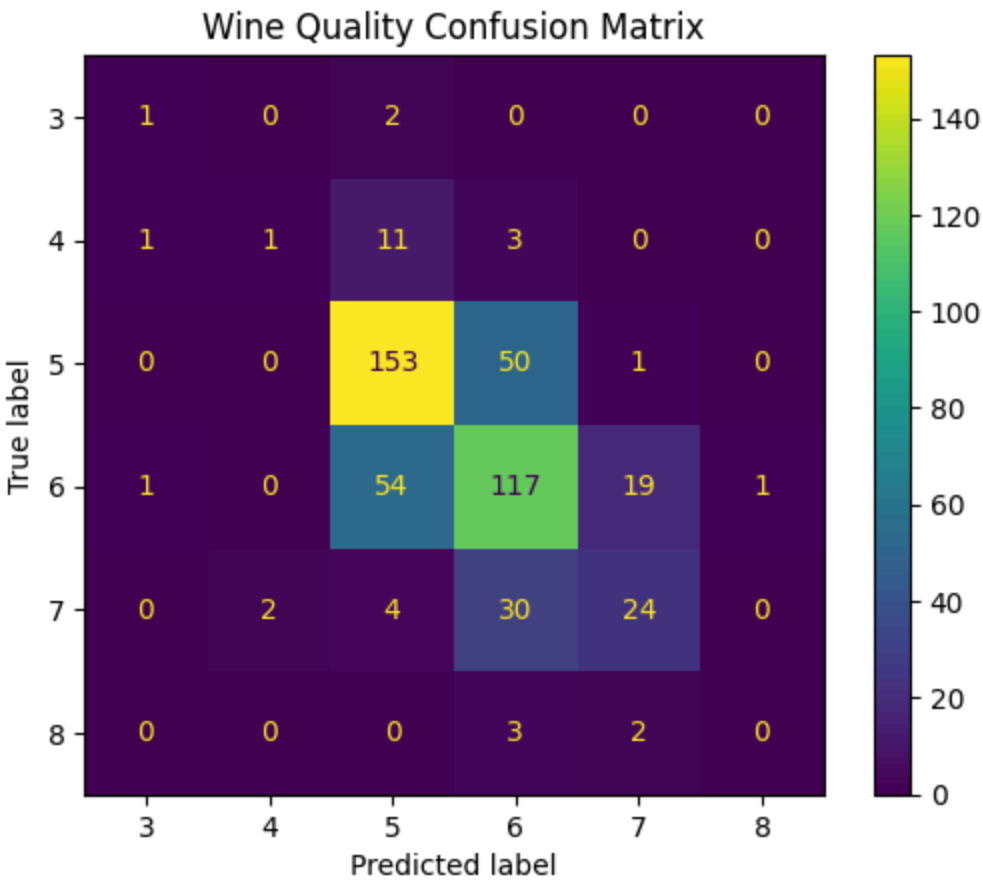
In [7]:
```
best_model = create_model(study.best_trial)
best_model.fit(X_train, y_train)
scores = cross_val_score(best_model, X_train, y_train, cv=5)

print("Best CV accuracy:", np.mean(scores))
```

```
Best CV accuracy: 0.6076993914157592
```

In [8]:
```
y_pred = best_model.predict(X_test)

acc = accuracy_score(y_test, y_pred)
f1 = f1_score(y_test, y_pred, average="weighted")

print("Test Accuracy:", acc)
print("Test F1-score:", f1)
```

```
Test Accuracy: 0.6166666666666667
Test F1-score: 0.6030130443514351
```

In [9]:
```
labels = np.unique(y)

cfm = confusion_matrix(y_test, y_pred, labels=labels)
cm_display = ConfusionMatrixDisplay(confusion_matrix=cfm,display_labels=labels)
cm_display.plot()
plt.title("Wine Quality Confusion Matrix")
plt.show()
```



In [10]:

```
#for letter b
def create_model(trial):
    model_type = trial.suggest_categorical('model_type',['SVR', 'KRR', 'GPR'])
    if model_type == 'SVR':
        kernel = trial.suggest_categorical('SVR-kernel',['linear','poly','rbf','sig
        C = trial.suggest_float('SVR-C',0.1, 100,log=True)
        gamma = trial.suggest_float('SVR-gamma',0.0001, 10,log=True)
        epsilon = trial.suggest_float('SVR-epsilon',0.001, 1,log=True)
        model = SVR(kernel=kernel,C=C,gamma=gamma,epsilon=epsilon)

    if model_type == 'KRR':
        alpha = trial.suggest_float('KRR-alpha',0.001, 10,log=True)
        kernel = trial.suggest_categorical('KRR-kernel',['linear','poly','rbf'])
        model = KernelRidge(alpha=alpha,kernel=kernel)

    if model_type == 'GPR':
        alpha = trial.suggest_float('GPR-alpha',1e-8, 1,log=True)
        kernel_type = trial.suggest_categorical('GPR-kernel',['rbf','matern','dot']
        if kernel_type == 'rbf':
            kernel = RBF()
        if kernel_type == 'matern':
            kernel = Matern()
        if kernel_type == 'dot':
            kernel = DotProduct()
        kernel = kernel + WhiteKernel(noise_level=1e-8, noise_level_bounds=(1e-10,
        model = GaussianProcessRegressor(kernel=kernel,alpha=alpha)

    if trial.should_prune():
```

```
        raise optuna.TrialPruned()

    return model
```

In [11]:
```
def objective(trial):
    model = create_model(trial)
    scores = cross_val_score(model,X_train,y_train,cv=5,scoring='neg_mean_squared_e
    return np.mean(scores)
```

In [12]:
```
start = time()

study = optuna.create_study(direction='maximize')
study.optimize(objective, n_trials=100)
best_params = study.best_params

print("\nBest parameters:", best_params)
```

```
[I 2026-03-06 19:39:37,299] A new study created in memory with name: no-name-6ce31d3
3-bcfa-4d62-b9b4-ae40b6a9a55d
[I 2026-03-06 19:39:37,422] Trial 0 finished with value: -0.4318622217351412 and par
ameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.17756642298491518, 'S
VR-gamma': 0.015884196133877343, 'SVR-epsilon': 0.2412331198618962}. Best is trial 0
with value: -0.4318622217351412.
[I 2026-03-06 19:39:37,487] Trial 1 finished with value: -32.6805670955947 and param
eters: {'model_type': 'KRR', 'KRR-alpha': 0.30957420636505517, 'KRR-kernel': 'linea
r'}. Best is trial 0 with value: -0.4318622217351412.
[I 2026-03-06 19:39:45,081] Trial 2 finished with value: -0.4367953786777761 and par
ameters: {'model_type': 'GPR', 'GPR-alpha': 5.849523898255266e-07, 'GPR-kernel': 'do
t'}. Best is trial 0 with value: -0.4318622217351412.
[I 2026-03-06 19:39:45,242] Trial 3 finished with value: -0.445122478990364 and para
meters: {'model_type': 'SVR', 'SVR-kernel': 'linear', 'SVR-C': 0.1030085154657493,
'SVR-gamma': 1.1977980788263356, 'SVR-epsilon': 0.0754322127904258}. Best is trial 0
with value: -0.4318622217351412.
[I 2026-03-06 19:39:51,095] Trial 4 finished with value: -0.43679537940248386 and pa
rameters: {'model_type': 'GPR', 'GPR-alpha': 1.0852678424175406e-05, 'GPR-kernel':
'dot'}. Best is trial 0 with value: -0.4318622217351412.
[I 2026-03-06 19:39:51,240] Trial 5 finished with value: -0.758964979450272 and para
meters: {'model_type': 'SVR', 'SVR-kernel': 'poly', 'SVR-C': 2.651535410541008, 'SVR
-gamma': 0.000884719077559974, 'SVR-epsilon': 0.03824481823803914}. Best is trial 0
with value: -0.4318622217351412.
[I 2026-03-06 19:39:51,297] Trial 6 finished with value: -32.68090421970072 and para
meters: {'model_type': 'KRR', 'KRR-alpha': 0.06890350013682683, 'KRR-kernel': 'linea
r'}. Best is trial 0 with value: -0.4318622217351412.
[I 2026-03-06 19:39:54,650] Trial 7 finished with value: -0.43678754340402454 and pa
rameters: {'model_type': 'GPR', 'GPR-alpha': 0.5321448999660388, 'GPR-kernel': 'do
t'}. Best is trial 0 with value: -0.4318622217351412.
[I 2026-03-06 19:40:00,126] Trial 8 finished with value: -0.43679537877016356 and pa
rameters: {'model_type': 'GPR', 'GPR-alpha': 7.414541719115101e-06, 'GPR-kernel': 'd
ot'}. Best is trial 0 with value: -0.4318622217351412.
/home/nomnom/miniconda3/envs/py37/lib/python3.7/site-packages/sklearn/gaussian_proce
ss/kernels.py:437: ConvergenceWarning: The optimal value found for dimension 0 of pa
rameter k1__sigma_0 is close to the specified upper bound 100000.0. Increasing the b
ound and calling fit again may find a better value.
  ConvergenceWarning,
[I 2026-03-06 19:40:07,473] Trial 9 finished with value: -0.4370874812482093 and par
ameters: {'model_type': 'GPR', 'GPR-alpha': 0.005587978842142669, 'GPR-kernel': 'do
t'}. Best is trial 0 with value: -0.4318622217351412.
[I 2026-03-06 19:40:07,527] Trial 10 finished with value: -0.5392252834785255 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.13282651094696593,
'SVR-gamma': 0.01801686475913448, 'SVR-epsilon': 0.9534576911552544}. Best is trial
0 with value: -0.4318622217351412.
[I 2026-03-06 19:40:08,808] Trial 11 finished with value: -0.6496695583542488 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 70.04407533020239, 'SV
R-gamma': 0.10102752115134553, 'SVR-epsilon': 0.002986522152666268}. Best is trial 0
with value: -0.4318622217351412.
[I 2026-03-06 19:40:16,742] Trial 12 finished with value: -0.4271292491154094 and pa
rameters: {'model_type': 'GPR', 'GPR-alpha': 0.18899599912639561, 'GPR-kernel': 'rb
f'}. Best is trial 12 with value: -0.4271292491154094.
[I 2026-03-06 19:40:16,833] Trial 13 finished with value: -0.5894707094473601 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'sigmoid', 'SVR-C': 2.654406504674895,
'SVR-gamma': 0.00015836097079894183, 'SVR-epsilon': 0.7236547678431733}. Best is tri
al 12 with value: -0.4271292491154094.
[I 2026-03-06 19:40:16,907] Trial 14 finished with value: -1.0781388406782648 and pa
```

```
rameters: {'model_type': 'KRR', 'KRR-alpha': 0.0010370622057778323, 'KRR-kernel': 'r
bf'}. Best is trial 12 with value: -0.4271292491154094.
[I 2026-03-06 19:40:24,162] Trial 15 finished with value: -0.43699164657494755 and p
arameters: {'model_type': 'GPR', 'GPR-alpha': 0.9031483966875149, 'GPR-kernel': 'rb
f'}. Best is trial 12 with value: -0.4271292491154094.
[I 2026-03-06 19:40:24,359] Trial 16 finished with value: -0.4304408598416926 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.5346257142751775, 'S
VR-gamma': 0.008285786804814845, 'SVR-epsilon': 0.11061619402874535}. Best is trial
12 with value: -0.4271292491154094.
[I 2026-03-06 19:40:26,801] Trial 17 finished with value: -2.0050241338972072 and pa
rameters: {'model_type': 'GPR', 'GPR-alpha': 0.0021230424066922137, 'GPR-kernel': 'r
bf'}. Best is trial 12 with value: -0.4271292491154094.
[I 2026-03-06 19:40:27,147] Trial 18 finished with value: -0.44786355428852564 and p
arameters: {'model_type': 'SVR', 'SVR-kernel': 'linear', 'SVR-C': 0.814838398627978
4, 'SVR-gamma': 7.1313986429064, 'SVR-epsilon': 0.0055529599523179715}. Best is tria
l 12 with value: -0.4271292491154094.
[I 2026-03-06 19:40:27,287] Trial 19 finished with value: -0.4665367475961772 and pa
rameters: {'model_type': 'KRR', 'KRR-alpha': 8.880225872299718, 'KRR-kernel': 'pol
y'}. Best is trial 12 with value: -0.4271292491154094.
/home/nomnom/miniconda3/envs/py37/lib/python3.7/site-packages/sklearn/gaussian_proce
ss/kernels.py:427: ConvergenceWarning: The optimal value found for dimension 0 of pa
rameter k2__noise_level is close to the specified lower bound 1e-10. Decreasing the
bound and calling fit again may find a better value.
  ConvergenceWarning,
/home/nomnom/miniconda3/envs/py37/lib/python3.7/site-packages/sklearn/gaussian_proce
ss/kernels.py:427: ConvergenceWarning: The optimal value found for dimension 0 of pa
rameter k2__noise_level is close to the specified lower bound 1e-10. Decreasing the
bound and calling fit again may find a better value.
  ConvergenceWarning,
/home/nomnom/miniconda3/envs/py37/lib/python3.7/site-packages/sklearn/gaussian_proce
ss/kernels.py:427: ConvergenceWarning: The optimal value found for dimension 0 of pa
rameter k2__noise_level is close to the specified lower bound 1e-10. Decreasing the
bound and calling fit again may find a better value.
  ConvergenceWarning,
/home/nomnom/miniconda3/envs/py37/lib/python3.7/site-packages/sklearn/gaussian_proce
ss/kernels.py:427: ConvergenceWarning: The optimal value found for dimension 0 of pa
rameter k2__noise_level is close to the specified lower bound 1e-10. Decreasing the
bound and calling fit again may find a better value.
  ConvergenceWarning,
/home/nomnom/miniconda3/envs/py37/lib/python3.7/site-packages/sklearn/gaussian_proce
ss/kernels.py:427: ConvergenceWarning: The optimal value found for dimension 0 of pa
rameter k2__noise_level is close to the specified lower bound 1e-10. Decreasing the
bound and calling fit again may find a better value.
  ConvergenceWarning,
[I 2026-03-06 19:40:36,382] Trial 20 finished with value: -0.699785083184059 and par
ameters: {'model_type': 'GPR', 'GPR-alpha': 4.152995065727066e-08, 'GPR-kernel': 'ma
tern'}. Best is trial 12 with value: -0.4271292491154094.
[I 2026-03-06 19:40:36,568] Trial 21 finished with value: -0.43022118456479375 and p
arameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.45244953256870307,
'SVR-gamma': 0.008439899523382914, 'SVR-epsilon': 0.17339948227849125}. Best is tria
l 12 with value: -0.4271292491154094.
[I 2026-03-06 19:40:36,734] Trial 22 finished with value: -0.43668001827258945 and p
arameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.7500381387278903,
'SVR-gamma': 0.0025987330666692537, 'SVR-epsilon': 0.1365792669318358}. Best is tria
l 12 with value: -0.4271292491154094.
[I 2026-03-06 19:40:36,934] Trial 23 finished with value: -0.41841138828798685 and p
```

```
arameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.6139676869910116,
'SVR-gamma': 0.15759215511022778, 'SVR-epsilon': 0.012146061316138053}. Best is tria
l 23 with value: -0.41841138828798685.
[I 2026-03-06 19:40:55,821] Trial 24 finished with value: -1.6506220557220437 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'poly', 'SVR-C': 19.56418821149448, 'S
VR-gamma': 0.16435029586140829, 'SVR-epsilon': 0.013949015045121332}. Best is trial
23 with value: -0.41841138828798685.
[I 2026-03-06 19:40:56,103] Trial 25 finished with value: -351.12742587675706 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'sigmoid', 'SVR-C': 0.404502352212662
2, 'SVR-gamma': 0.40844366761111545, 'SVR-epsilon': 0.013405574764586205}. Best is t
rial 23 with value: -0.41841138828798685.
[I 2026-03-06 19:40:56,362] Trial 26 finished with value: -0.47879060284938957 and p
arameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 7.24558680985745, 'SV
R-gamma': 1.8618585113730528, 'SVR-epsilon': 0.019890763961074035}. Best is trial 23
with value: -0.41841138828798685.
[I 2026-03-06 19:40:56,579] Trial 27 finished with value: -0.42537503988638675 and p
arameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 1.442806496505952, 'S
VR-gamma': 0.057410787272568875, 'SVR-epsilon': 0.0013964219494760736}. Best is tria
l 23 with value: -0.41841138828798685.
[I 2026-03-06 19:40:56,744] Trial 28 finished with value: -3.540525741746272 and par
ameters: {'model_type': 'KRR', 'KRR-alpha': 0.0011800063467465242, 'KRR-kernel': 'po
ly'}. Best is trial 23 with value: -0.41841138828798685.
[I 2026-03-06 19:40:57,871] Trial 29 finished with value: -1.7289117498630517 and pa
rameters: {'model_type': 'GPR', 'GPR-alpha': 0.007277756143878493, 'GPR-kernel': 'rb
f'}. Best is trial 23 with value: -0.41841138828798685.
[I 2026-03-06 19:40:58,114] Trial 30 finished with value: -0.4272945021746268 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 1.700055481308629, 'SV
R-gamma': 0.09147442139895774, 'SVR-epsilon': 0.004559952354116216}. Best is trial 2
3 with value: -0.41841138828798685.
[I 2026-03-06 19:40:58,340] Trial 31 finished with value: -0.4305242904169468 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 2.157124521316091, 'SV
R-gamma': 0.07435042240266766, 'SVR-epsilon': 0.0012194262902486999}. Best is trial
23 with value: -0.41841138828798685.
[I 2026-03-06 19:40:58,549] Trial 32 finished with value: -0.4231887231931788 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 1.3367292672540498, 'S
VR-gamma': 0.05022477679081473, 'SVR-epsilon': 0.0011133863055737795}. Best is trial
23 with value: -0.41841138828798685.
[I 2026-03-06 19:40:58,843] Trial 33 finished with value: -0.46875744968754757 and p
arameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 5.1444101316528, 'SVR
-gamma': 0.4559033893698719, 'SVR-epsilon': 0.0012475931657524575}. Best is trial 23
with value: -0.41841138828798685.
[I 2026-03-06 19:40:59,007] Trial 34 finished with value: -0.5470796994409846 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'poly', 'SVR-C': 1.4067114990132203,
'SVR-gamma': 0.04464752976855742, 'SVR-epsilon': 0.001981543540481645}. Best is tria
l 23 with value: -0.41841138828798685.
[I 2026-03-06 19:40:59,199] Trial 35 finished with value: -0.4477442509222153 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'linear', 'SVR-C': 0.2370679391190608
5, 'SVR-gamma': 0.3046931159752541, 'SVR-epsilon': 0.00605703861895973}. Best is tri
al 23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:01,817] Trial 36 finished with value: -0.6957101398496632 and pa
rameters: {'model_type': 'GPR', 'GPR-alpha': 0.0004588932772515572, 'GPR-kernel': 'm
atern'}. Best is trial 23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:01,895] Trial 37 finished with value: -1.9436941629577178 and pa
rameters: {'model_type': 'KRR', 'KRR-alpha': 9.153875438123638, 'KRR-kernel': 'rb
f'}. Best is trial 23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:02,145] Trial 38 finished with value: -0.4466928191195133 and pa
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rameters: {'model_type': 'SVR', 'SVR-kernel': 'sigmoid', 'SVR-C': 7.475803018269428,
'SVR-gamma': 0.0035892326642518817, 'SVR-epsilon': 0.0024176531927758405}. Best is t
rial 23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:11,400] Trial 39 finished with value: -0.5561874384825051 and pa
rameters: {'model_type': 'GPR', 'GPR-alpha': 0.0779254940724737, 'GPR-kernel': 'rb
f'}. Best is trial 23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:11,626] Trial 40 finished with value: -0.4248118660780228 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 1.0984465352994217, 'S
VR-gamma': 0.03052500582319371, 'SVR-epsilon': 0.0011401410635733073}. Best is trial
23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:11,825] Trial 41 finished with value: -0.42532063471840154 and p
arameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.8916260667373167,
'SVR-gamma': 0.03177895323620638, 'SVR-epsilon': 0.001159312885962281}. Best is tria
l 23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:12,025] Trial 42 finished with value: -0.42265746554117295 and p
arameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 1.0103199868043522,
'SVR-gamma': 0.03562102016005119, 'SVR-epsilon': 0.0010185218921599116}. Best is tri
al 23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:12,218] Trial 43 finished with value: -0.42611796223159965 and p
arameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.9155033946200978,
'SVR-gamma': 0.026576001180245393, 'SVR-epsilon': 0.0010371116839043753}. Best is tr
ial 23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:12,418] Trial 44 finished with value: -0.42500693378161813 and p
arameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 1.079219474298794, 'S
VR-gamma': 0.030129784460728195, 'SVR-epsilon': 0.002729607493129763}. Best is trial
23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:12,611] Trial 45 finished with value: -0.42404108876930735 and p
arameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.3315332206114868,
'SVR-gamma': 0.1817456820764001, 'SVR-epsilon': 0.00308150200907211}. Best is trial
23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:12,795] Trial 46 finished with value: -0.4257176693982941 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.2670050562036613, 'S
VR-gamma': 0.1489652301548639, 'SVR-epsilon': 0.004167747471001525}. Best is trial 2
3 with value: -0.41841138828798685.
[I 2026-03-06 19:41:12,986] Trial 47 finished with value: -0.4225033586372959 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.39507277887178294,
'SVR-gamma': 0.18044537419755294, 'SVR-epsilon': 0.008811475123404805}. Best is tria
l 23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:13,177] Trial 48 finished with value: -0.4583369297416892 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.41932258474477874,
'SVR-gamma': 0.737172860713233, 'SVR-epsilon': 0.00738227914554806}. Best is trial 2
3 with value: -0.41841138828798685.
[I 2026-03-06 19:41:13,358] Trial 49 finished with value: -0.43346854444642596 and p
arameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.22390174601716956,
'SVR-gamma': 0.21784277121637638, 'SVR-epsilon': 0.010226458093937144}. Best is tria
l 23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:13,604] Trial 50 finished with value: -0.5732557433915442 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.32217570243902294,
'SVR-gamma': 2.9116349837507314, 'SVR-epsilon': 0.028557739181305542}. Best is trial
23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:13,783] Trial 51 finished with value: -0.43221620196524463 and p
arameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.5995738065111205,
'SVR-gamma': 0.012109998429378495, 'SVR-epsilon': 0.001908636505306892}. Best is tri
al 23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:13,982] Trial 52 finished with value: -0.42086745840501505 and p
arameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.6109896625611283,
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'SVR-gamma': 0.06834846970795062, 'SVR-epsilon': 0.0035863722924706297}. Best is tri
al 23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:14,185] Trial 53 finished with value: -0.41907185966105465 and p
arameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.6042144184516117,
'SVR-gamma': 0.1255632884914329, 'SVR-epsilon': 0.0034019128219568285}. Best is tria
l 23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:14,360] Trial 54 finished with value: -0.4475542053912168 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'linear', 'SVR-C': 0.1744077178143301
8, 'SVR-gamma': 0.07590176776347196, 'SVR-epsilon': 0.007738771481672759}. Best is t
rial 23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:14,439] Trial 55 finished with value: -0.645202790025787 and par
ameters: {'model_type': 'KRR', 'KRR-alpha': 0.024485261897467785, 'KRR-kernel': 'rb
f'}. Best is trial 23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:14,680] Trial 56 finished with value: -0.44243557830281616 and p
arameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.5456764628879631,
'SVR-gamma': 0.5736516956455924, 'SVR-epsilon': 0.0017808287995720181}. Best is tria
l 23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:15,708] Trial 57 finished with value: -1.268341372918361 and par
ameters: {'model_type': 'SVR', 'SVR-kernel': 'poly', 'SVR-C': 4.0482294538721355, 'S
VR-gamma': 0.1152808664444541, 'SVR-epsilon': 0.0035221771690137015}. Best is trial
23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:15,882] Trial 58 finished with value: -0.43038691683764085 and p
arameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.6619070271816936,
'SVR-gamma': 0.016931115342463263, 'SVR-epsilon': 0.03925641504911264}. Best is tria
l 23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:16,111] Trial 59 finished with value: -0.5074768802138356 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'sigmoid', 'SVR-C': 0.1025278048896547
1, 'SVR-gamma': 0.05436529745554302, 'SVR-epsilon': 0.009892498785009553}. Best is t
rial 23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:16,362] Trial 60 finished with value: -0.4462040422891659 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 2.227019857561006, 'SV
R-gamma': 0.27165823814530854, 'SVR-epsilon': 0.004772002874384302}. Best is trial 2
3 with value: -0.41841138828798685.
[I 2026-03-06 19:41:16,551] Trial 61 finished with value: -0.4231122192838025 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.3401316813781697, 'S
VR-gamma': 0.16739339391381036, 'SVR-epsilon': 0.0029487294871171385}. Best is trial
23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:16,731] Trial 62 finished with value: -0.4329771078414126 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.1643027715456801, 'S
VR-gamma': 0.11693514223625169, 'SVR-epsilon': 0.0019768314032146046}. Best is trial
23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:16,928] Trial 63 finished with value: -0.4222033188354845 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.6030643459178456, 'S
VR-gamma': 0.05335859954908652, 'SVR-epsilon': 0.0033994684152322194}. Best is trial
23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:17,120] Trial 64 finished with value: -0.4596905574717723 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.4650377608770417, 'S
VR-gamma': 0.8224949500753644, 'SVR-epsilon': 0.0033019748470185}. Best is trial 23
with value: -0.41841138828798685.
[I 2026-03-06 19:41:17,175] Trial 65 finished with value: -32.67984568077379 and par
ameters: {'model_type': 'KRR', 'KRR-alpha': 0.8277413522196085, 'KRR-kernel': 'linea
r'}. Best is trial 23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:17,333] Trial 66 finished with value: -0.42994695816326506 and p
arameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.6474086963176445,
'SVR-gamma': 0.2573770030955486, 'SVR-epsilon': 0.4311640688753671}. Best is trial 2
3 with value: -0.41841138828798685.
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[I 2026-03-06 19:41:17,535] Trial 67 finished with value: -0.421220644089716 and par
ameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.3782369234443914, 'SV
R-gamma': 0.13184551684917353, 'SVR-epsilon': 0.00622392338295643}. Best is trial 23
with value: -0.41841138828798685.
[I 2026-03-06 19:41:17,819] Trial 68 finished with value: -0.4479372711054899 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'linear', 'SVR-C': 0.7079299918839057,
'SVR-gamma': 0.007119941369806435, 'SVR-epsilon': 0.018140551718656014}. Best is tri
al 23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:18,031] Trial 69 finished with value: -0.625468389993151 and par
ameters: {'model_type': 'SVR', 'SVR-kernel': 'poly', 'SVR-C': 0.4391601858423401, 'S
VR-gamma': 0.09727448731550661, 'SVR-epsilon': 0.006201051698406713}. Best is trial
23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:18,198] Trial 70 finished with value: -0.6716661664343178 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.27036728013117806,
'SVR-gamma': 0.00020448730620419093, 'SVR-epsilon': 0.009744553210766295}. Best is t
rial 23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:18,396] Trial 71 finished with value: -0.4223178874316832 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.39946141000277663,
'SVR-gamma': 0.17960694024871865, 'SVR-epsilon': 0.004393569950347768}. Best is tria
l 23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:18,591] Trial 72 finished with value: -0.4331314000255545 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.5154885110056593, 'S
VR-gamma': 0.3699892636874791, 'SVR-epsilon': 0.004326670155281303}. Best is trial 2
3 with value: -0.41841138828798685.
[I 2026-03-06 19:41:18,786] Trial 73 finished with value: -0.42122166795005755 and p
arameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.8275930528514434,
'SVR-gamma': 0.07490002711711291, 'SVR-epsilon': 0.013115150483161484}. Best is tria
l 23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:18,970] Trial 74 finished with value: -0.4224867449669059 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.3714484002588027, 'S
VR-gamma': 0.07050638844421, 'SVR-epsilon': 0.01668987918917663}. Best is trial 23 w
ith value: -0.41841138828798685.
[I 2026-03-06 19:41:19,202] Trial 75 finished with value: -0.4213184325543556 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.8003774753804516, 'S
VR-gamma': 0.06871470975160532, 'SVR-epsilon': 0.014632689813742565}. Best is trial
23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:19,458] Trial 76 finished with value: -75.50505009353697 and par
ameters: {'model_type': 'SVR', 'SVR-kernel': 'sigmoid', 'SVR-C': 0.797262826238959,
'SVR-gamma': 0.10574937571524658, 'SVR-epsilon': 0.026075667948149125}. Best is tria
l 23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:19,590] Trial 77 finished with value: -1.5502327937909324 and pa
rameters: {'model_type': 'KRR', 'KRR-alpha': 0.009168885171863858, 'KRR-kernel': 'po
ly'}. Best is trial 23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:20,491] Trial 78 finished with value: -0.5357535644472364 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 57.13999850387994, 'SV
R-gamma': 0.05612898138785384, 'SVR-epsilon': 0.013352043539388115}. Best is trial 2
3 with value: -0.41841138828798685.
/home/nomnom/miniconda3/envs/py37/lib/python3.7/site-packages/sklearn/gaussian_proce
ss/kernels.py:427: ConvergenceWarning: The optimal value found for dimension 0 of pa
rameter k2__noise_level is close to the specified lower bound 1e-10. Decreasing the
bound and calling fit again may find a better value.
  ConvergenceWarning,
/home/nomnom/miniconda3/envs/py37/lib/python3.7/site-packages/sklearn/gaussian_proce
ss/kernels.py:427: ConvergenceWarning: The optimal value found for dimension 0 of pa
rameter k2__noise_level is close to the specified lower bound 1e-10. Decreasing the
bound and calling fit again may find a better value.
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  ConvergenceWarning,
/home/nomnom/miniconda3/envs/py37/lib/python3.7/site-packages/sklearn/gaussian_proce
ss/kernels.py:427: ConvergenceWarning: The optimal value found for dimension 0 of pa
rameter k2__noise_level is close to the specified lower bound 1e-10. Decreasing the
bound and calling fit again may find a better value.
  ConvergenceWarning,
/home/nomnom/miniconda3/envs/py37/lib/python3.7/site-packages/sklearn/gaussian_proce
ss/kernels.py:427: ConvergenceWarning: The optimal value found for dimension 0 of pa
rameter k2__noise_level is close to the specified lower bound 1e-10. Decreasing the
bound and calling fit again may find a better value.
  ConvergenceWarning,
/home/nomnom/miniconda3/envs/py37/lib/python3.7/site-packages/sklearn/gaussian_proce
ss/kernels.py:427: ConvergenceWarning: The optimal value found for dimension 0 of pa
rameter k2__noise_level is close to the specified lower bound 1e-10. Decreasing the
bound and calling fit again may find a better value.
  ConvergenceWarning,
[I 2026-03-06 19:41:27,058] Trial 79 finished with value: -0.6997854534935762 and pa
rameters: {'model_type': 'GPR', 'GPR-alpha': 1.0493735261605583e-08, 'GPR-kernel':
'matern'}. Best is trial 23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:27,270] Trial 80 finished with value: -0.4284266586223011 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 1.8392603156220013, 'S
VR-gamma': 0.020261699303386088, 'SVR-epsilon': 0.05470576318866886}. Best is trial
23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:27,469] Trial 81 finished with value: -0.42049492169352265 and p
arameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.5475203168161569,
'SVR-gamma': 0.08674971413033684, 'SVR-epsilon': 0.012232439873325923}. Best is tria
l 23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:27,662] Trial 82 finished with value: -0.4205396302079987 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.5712832179224221, 'S
VR-gamma': 0.07997794934721099, 'SVR-epsilon': 0.013916141619460509}. Best is trial
23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:27,860] Trial 83 finished with value: -0.42389828273198765 and p
arameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.5903734621472776,
'SVR-gamma': 0.044006574731618855, 'SVR-epsilon': 0.013011939735153057}. Best is tri
al 23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:28,063] Trial 84 finished with value: -0.42058670210831817 and p
arameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 1.1148469381382717,
'SVR-gamma': 0.07712862370840456, 'SVR-epsilon': 0.024207160737359137}. Best is tria
l 23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:28,254] Trial 85 finished with value: -0.4201234239956932 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.8436369768936786, 'S
VR-gamma': 0.07412122821705065, 'SVR-epsilon': 0.022168607054083394}. Best is trial
23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:28,457] Trial 86 finished with value: -0.4219612828891982 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 1.2123741729663473, 'S
VR-gamma': 0.11903700907329758, 'SVR-epsilon': 0.023675886776834243}. Best is trial
23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:28,869] Trial 87 finished with value: -0.44703186536655276 and p
arameters: {'model_type': 'SVR', 'SVR-kernel': 'linear', 'SVR-C': 1.439184961975240
8, 'SVR-gamma': 0.08220429050173315, 'SVR-epsilon': 0.03653710621592445}. Best is tr
ial 23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:29,072] Trial 88 finished with value: -0.4247282290242156 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 1.0209024049648323, 'S
VR-gamma': 0.2849330695777318, 'SVR-epsilon': 0.02141491607150276}. Best is trial 23
with value: -0.41841138828798685.
[I 2026-03-06 19:41:29,207] Trial 89 finished with value: -0.5498726674443773 and pa
```

```
rameters: {'model_type': 'SVR', 'SVR-kernel': 'poly', 'SVR-C': 0.8396489201054863,
'SVR-gamma': 0.022584227748413455, 'SVR-epsilon': 0.005887062648937723}. Best is tri
al 23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:31,313] Trial 90 finished with value: -0.6990083173650712 and pa
rameters: {'model_type': 'GPR', 'GPR-alpha': 8.261765858183048e-05, 'GPR-kernel': 'm
atern'}. Best is trial 23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:31,543] Trial 91 finished with value: -0.4209146808214331 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.7993308446588262, 'S
VR-gamma': 0.07280007534110507, 'SVR-epsilon': 0.015747252490853996}. Best is trial
23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:31,727] Trial 92 finished with value: -0.42651659820332355 and p
arameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.5180883803374048,
'SVR-gamma': 0.03788653426936953, 'SVR-epsilon': 0.012354651571281123}. Best is tria
l 23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:31,925] Trial 93 finished with value: -0.4188078438709465 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.7436920092627679, 'S
VR-gamma': 0.14485908423361424, 'SVR-epsilon': 0.018030464696875827}. Best is trial
23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:32,129] Trial 94 finished with value: -0.4209321201724297 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 1.0996541865150409, 'S
VR-gamma': 0.12843873958449906, 'SVR-epsilon': 0.019758820111107706}. Best is trial
23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:32,338] Trial 95 finished with value: -0.4289917896109345 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 1.653124593768258, 'SV
R-gamma': 0.2041450342755358, 'SVR-epsilon': 0.034407680043870105}. Best is trial 23
with value: -0.41841138828798685.
[I 2026-03-06 19:41:32,542] Trial 96 finished with value: -0.4216325110376319 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 1.1785444366607547, 'S
VR-gamma': 0.13164237141947854, 'SVR-epsilon': 0.019198568248736943}. Best is trial
23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:32,601] Trial 97 finished with value: -32.6800468761945 and para
meters: {'model_type': 'KRR', 'KRR-alpha': 0.6827971536298951, 'KRR-kernel': 'linea
r'}. Best is trial 23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:32,912] Trial 98 finished with value: -15435.804561193969 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'sigmoid', 'SVR-C': 2.923337065750817
8, 'SVR-gamma': 0.3647730926566899, 'SVR-epsilon': 0.0656152629312888}. Best is tria
l 23 with value: -0.41841138828798685.
[I 2026-03-06 19:41:33,102] Trial 99 finished with value: -0.4211631020847938 and pa
rameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.9713969726832496, 'S
VR-gamma': 0.039742803431174686, 'SVR-epsilon': 0.04307947818804048}. Best is trial
23 with value: -0.41841138828798685.
Best parameters: {'model_type': 'SVR', 'SVR-kernel': 'rbf', 'SVR-C': 0.6139676869910
116, 'SVR-gamma': 0.15759215511022778, 'SVR-epsilon': 0.012146061316138053}
```

In [13]:

```
best_model = create_model(study.best_trial)
best_model.fit(X_train, y_train)
scores = cross_val_score(best_model,X_train,y_train,cv=5,scoring='neg_mean_squared_
print("Best CV MSE:", -np.mean(scores))
```

```
Best CV MSE: 0.41841138828798685
```

In [14]:

```
y_pred = best_model.predict(X_test)

mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)
mad = mean_absolute_error(y_test, y_pred)
```

```
print("Test MSE:", mse)
print("Test R2:", r2)
print("Test MAD:", mad)
# MAD value was around 0.44 compared to the paper which had a value of 0.46, sugges
#the model in the paper.
```

```
Test MSE: 0.4017703147860965
Test R2: 0.3802987077709349
Test MAD: 0.4415561616176748
```

In [ ]:

```
#If treated as classification which predicted discrete values for quality of wine,
# F1 score of (0.603). If treated as regression, it will predict continouos value f
#estimates. My values were MSE = 0.4018 and MAD = 0.4416, suggesting that the predi
# a half point away from actual wine rating.

# in general classification is easier if we need to label things to categories, whi
# differences in values, which can be better for detailed analysis.
```